Industrial Zone Location in Turkey: A Spatial Analysis of Organized Industrial Zones

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1 Data and Zone Construction

I selected Turkey because its Organized Industrial Zones (OSBs, *Organize Sanayi Bölgesi*) are governed by a dedicated legal framework (Law No. 4562), centrally registered by OSBÜK, and well-documented. The OSBÜK registry contains 297 active zones with names, provinces, and website addresses. Zones were geocoded using OpenStreetMap via `tidygeocoder`, querying each zone by name and province. Of 297 zones, 116 were matched (39.1%), covering 54 of 81 provinces. Unmatched zones were dropped, likely introducing a bias toward larger and better-documented zones. Geocoded points were spatially joined to GADM level-2 district boundaries, establishing the district as the unit of analysis.

2 Additional Spatial and Economic Data

All spatial objects use WGS 84 (EPSG:4326); distances are in kilometers. Three variables were constructed at the district level. **Distance to nearest motorway or trunk road** was calculated from each district centroid to the nearest road segment using HOT OSM Turkey road data (April 2026). **Night light intensity** was extracted as mean radiance from the VNL v2 2025 annual masked composite (Earth Observation Group, Colorado School of Mines) using `exact_extract()`. The masked composite removes ephemeral lights such as fires. **Socioeconomic status score** is a district-level index from TUIK (2023), computed as 50% education, 30% income, and 20% occupation score, ranging 0–300. Scores were merged by province-district name after normalizing Turkish special characters on both sides of the join. GADM uses pre-2013 administrative boundaries while TUIK reflects post-2013 metropolitan reforms; mismatches were resolved manually via a name crosswalk.

3 Zone vs. Non-Zone Locations

The unit of analysis is the GADM level-2 district — Turkey's second administrative tier below provinces, of which there are 973. Districts are the appropriate unit because OSBs are sub-provincial investments: comparing zone provinces to non-zone provinces would conflate the zone effect with the broader regional advantages that likely drove zone placement in the first place, making it an apples-to-oranges comparison. Staying within provinces and comparing adjacent districts helps isolate the zone as a more localized shock.

The goal is descriptive: do districts hosting an OSB differ from nearby districts without one in infrastructure access and economic activity? The preferred control group consists of districts sharing a border with a zone district (394 non-zone, 88 zone districts). Zone districts are explicitly excluded from the neighbor list. This geographically restricted comparison is more homogeneous than a province-wide one (CV of night lights: 2.59 vs. 2.87). All distance measures use district centroids for comparability across zone and non-zone districts.

Table 1 shows zone districts are on average 4.48 km from a motorway versus 10.14 km for adjacent non-zone districts, and have higher mean night light intensity (6.10 vs. 4.30) and higher mean SES scores (129.84 vs. 114.95). Cross-sectional regressions confirm zone districts score 13.62 points higher on SES within province after controlling for province fixed effects. The night lights coefficient (1.77) is marginally significant at the 10% level within province, suggesting the raw difference is partly explained by broader regional patterns rather than zone presence alone.

4 Dynamic and Causal Extension

Establishment years. Years were recovered by scraping individual OSB websites using `rvest`, checking homepages and six Turkish subpages per site. Years below 1961 — the founding year of Turkey's first OSB in Bursa — were excluded after manual inspection confirmed they reflected sector history. Years were recovered for 54 of 116 zones (46.6%); of these, only 9 opened after 2012, the start of VIIRS coverage.

Proposed design. I propose a staggered difference-in-differences using the Callaway-Sant'Anna (2021) estimator, comparing districts that received a zone in year t to adjacent never-treated districts, with district and year fixed effects and standard errors clustered at the district level. The outcome would be annual night light radiance. The key assumption is parallel trends; geographic proximity of adjacent districts makes this more plausible than a national comparison. A spatial extension would replace the binary treatment with a distance-decay measure to test for spillovers across district borders, using a contiguity-based spatial weights matrix.

Limitations and Next Steps. The analysis is cross-sectional: without panel data, zone effects cannot be separated from pre-existing district differences. Geocoding covers only 116 of 297 zones, likely overrepresenting larger and better-documented OSBs; the remaining zones could be recovered via manual geocoding using the address field in the OSBÜK registry. Establishment year coverage stands at 46.6%, and may reflect legal founding rather than operational start. The district-level comparison uses centroid-to-road distances throughout; exact zone coordinates are retained in `zones.csv` but not used here. VIIRS annual composites (2012–2023) and DMSP-OLS composites (1992–2013) were not downloaded, so a before-after design is not currently feasible. A panel design exploiting variation in establishment years would allow difference-in-differences or event study estimates of zone effects — feasible once night light panel data and scraping coverage are completed.

AI Disclosure

I used Claude for debugging, summarizing, and organizing. All analytical decisions are my own.